Explainable Bittergourd Freshness Detection using MobileNetV3 and GradCAM

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***Abstract*—Accurate detection of fruit freshness is important for reducing food waste, improving food safety, and enhancing post-harvest management. Traditional methods of identifying spoiled fruits are mostly manual, time-consuming, and may not provide consistent results. This study presents an explainable deep learning approach for classifying fresh and spoiled bitter gourd fruits using MobileNetV3 and GradCAM. MobileNetV3 was selected for its lightweight structure and suitability for low-resource and mobile-based applications, while GradCAM was integrated to provide visual explanations of the model's decisions by highlighting the most influential image regions. The proposed model achieved 100% classification accuracy, demonstrating its potential for efficient and transparent post-harvest quality monitoring.**

***Index Terms*—Freshness detection, Deep Learning, Fruit Classification.**

I. Introduction

Post-harvest management is a crucial part of agricultural activity aimed at maximizing the quality and shelf life of agricultural products, particularly fruits and vegetables, ensuring that they reach the market in the best possible condition [1]. If products are not handled correctly, they can lose quality, taste, flavour, texture, and nutrients, resulting in economic losses for producers and lower-quality products for consumers [2]. The global demand for food is expected to increase by 35% to 56% by 2050 compared to 2010, while global warming is expected to reduce crop frequency, complicate water supply, and diminish harvesting or planting time windows [3]. In the case of fruits and vegetables, it is estimated that worldwide, more than 50% of the yield is lost along the supply chain [3], with rotten crops being a primary contributing factor. The use of automated systems for detecting rotten fruits, unlike conventional manual classification methods, can provide a cost-effective and efficient solution, as accurately classifying rotten and fresh fruits is a pivotal factor in improving supply chain efficiency and reducing food waste [3].

Fruit classification has recently become a highly required task [1], and advances in deep learning, particularly Convolutional Neural Networks (CNNs), have created new opportunities for identifying and classifying fruits based on visual characteristics such as shape, size, colour, and texture [4]. Bitter gourd is an important plant of the Cucurbitaceae family with high nutritional content and is used in the treatment of many diseases [5] [3]; however, automated methods for assessing its freshness remain underexplored. MobileNetV3, designed for low-resource devices, enables highly accurate classification while conserving computational resources, making it suitable for mobile and edge applications [4]. However, the black-box nature of deep learning models raises concerns about transparency in food safety applications. To address this, GradCAM was integrated to provide visual explanations of the model's decisions by highlighting the image regions most influential to each prediction. This paper therefore presents an explainable deep learning approach for bitter gourd freshness detection using MobileNetV3 and GradCAM, contributing a transparent and efficient classification system suitable for post-harvest quality monitoring applications.

II. Proposed System

The proposed system is an image-based classification pipeline for detecting bitter gourd freshness using MobileNetV3 and GradCAM, as illustrated in Fig. 1. The pipeline consists of four stages: data preparation, model training, evaluation, and explainability.

A dataset of fresh and rotten bitter gourd images was sourced from Kaggle. All images were resized to 224×224 pixels and normalised using ImageNet statistics (mean $=$ $[0.485, 0.456, 0.406]$, std $=$ $[0.229, 0.224, 0.225]$). Training augmentation included random horizontal flipping ($p = 0.5$), rotation (± 15), colour jitter, and random grayscale ($p = 0.1$). The dataset was split into 80% training and 20% testing with stratified sampling to preserve class balance.

MobileNetV3-Small pretrained on ImageNet-1K was adopted using transfer learning. The final fully-connected layer was replaced with a linear layer mapping to two output classes: fresh and rotten. The model was optimised using Adam ($lr = 1 \times 10^{-4}$, weight decay $= 1 \times 10^{-4}$) with CrossEntropyLoss and a ReduceLROnPlateau scheduler (patience$= 3$, factor$= 0.1$) over 10 epochs with batch size 32, executed on a Google Colab T4 GPU.

GradCAM was applied to the final convolutional block of MobileNetV3 to generate class activation maps highlighting the image regions most influential to each prediction.

III. Results and discusion

The proposed MobileNetV3-based system was evaluated on a test set of 40 bitter gourd images (20 fresh, 20 rotten). Figs. 2 and 3 present the confusion matrix, and GradCAM visualisations respectively.

The model achieved a test accuracy of 100%, with precision, recall, and F1-score of 1.00 across both classes. The confusion matrix in Fig. 2 confirms perfect classification, with all 20

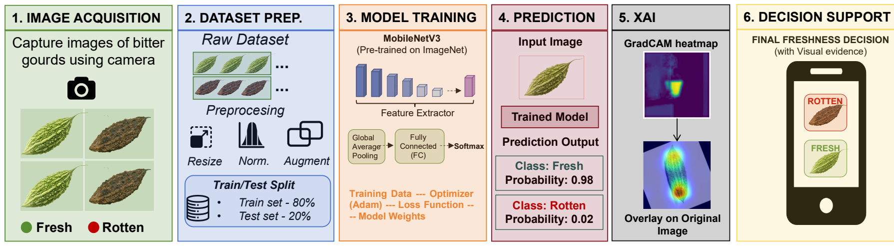

Fig. 1. Overview of the proposed bitter gourd freshness detection pipeline, comprising image preprocessing, MobileNetV3 transfer learning-based classification, and GradCAM explainability.

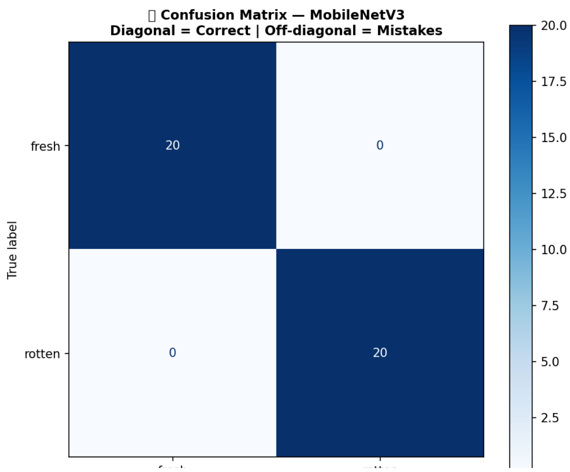

Fig. 2. Confusion matrix for MobileNetV3 on the bitter gourd test set. All 20 fresh and 20 rotten samples are correctly classified, yielding zero false positives and zero false negatives.

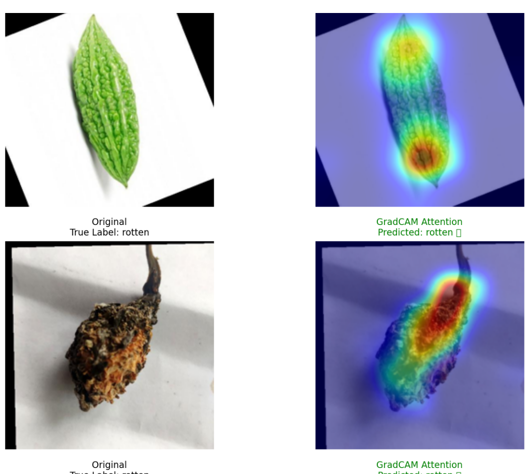

Fig. 3. GradCAM attention maps for a fresh bitter gourd (top) and a rotten bitter gourd (bottom). Warm colours (red/yellow) indicate regions of highest model attention. The model correctly focuses on the fruit body for the fresh sample and on the decayed stem and tissue for the rotten sample.

fresh and 20 rotten samples correctly identified and zero misclassifications in either direction.

Fig. 3 presents GradCAM attention maps for representative fresh and rotten bitter gourd samples. For the fresh sample, the model's attention is concentrated on the central body of the fruit, with the highest activation (red region) localised at the mid-section, consistent with the characteristic texture and colour of healthy bitter gourd skin. For the rotten sample, attention is focused on the decayed stem and deteriorated body tissue, precisely the regions exhibiting visible spoilage cues. In both cases, the model correctly attended to semantically meaningful regions of the fruit rather than the background, demonstrating that the classification decisions are grounded in visually relevant features and not spurious correlations.

IV. Conclusion

This paper presented an explainable deep learning approach for bitter gourd freshness detection using MobileNetV3 and GradCAM, achieving 100% classification accuracy on the test set. GradCAM visualisations confirmed that model decisions are grounded in semantically meaningful fruit regions, addressing transparency concerns in food safety applications. Future work will explore larger datasets, multi-class freshness grading, and edge device deployment for real-time post-harvest quality monitoring.

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